

Deep Learning on sPHENIX Calorimeter Waveform Fits



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Abstract

The sPHENIX experiment at RHIC is a state-of-the-art detector designed to explore the inner workings of the Quark-Gluon Plasma (QGP) produced in heavy ion collisions. To achieve its science goals the sPHENIX detector employs high-precision tracking and calorimetry and will take an unprecedented amount of minimum bias data. The sPHENIX calorimetry consists of one electromagnetic (EMCAL) and two hadronic calorimeters; the focus of this project is the EMCAL. The waveform digitized signals from the calorimeters are processed by a template-based fit, which extracts the energy, time and pedestal for each calorimeter tower. Due to the large number of EMCAL towers, this procedure is the most time-consuming step in the calorimeter data reconstruction. Therefore, we investigated a machine learning algorithm as an alternative technique for the fit-based feature extraction. The machine learning model is built using Keras, a high-level TensorFlow API, within Spyder IDE. This algorithm may provide quicker and more precise signal feature extraction than the current fit based approach.

Introduction

The sPHENIX detector (Figure 1) is a cutting-edge jet detector, succeeding the PHENIX detector at RHIC in IP8. The main physics goal of sPHENIX is to further study the properties of the quark-gluon plasma (QGP). By measuring jets produced by different quarks, the effect of color charge and mass on the energy loss of quarks in the QGP can be inferred. To satisfy its physics goals, sPHENIX will take data for two years: p+p in 2024 and a large Au+Au run in 2025. sPHENIX is designed to take unprecedented amounts of minimum bias data at a high rate of 15kHz. This will result in a 200PB dataset for the 2025 Au+Au run containing about 100B events. The processing of this data poses a unique challenge, so new avenues to efficiently reconstruct the data need to be investigated. Machine learning might provide a novel approach to solve common computing problems faster than current techniques.

Background

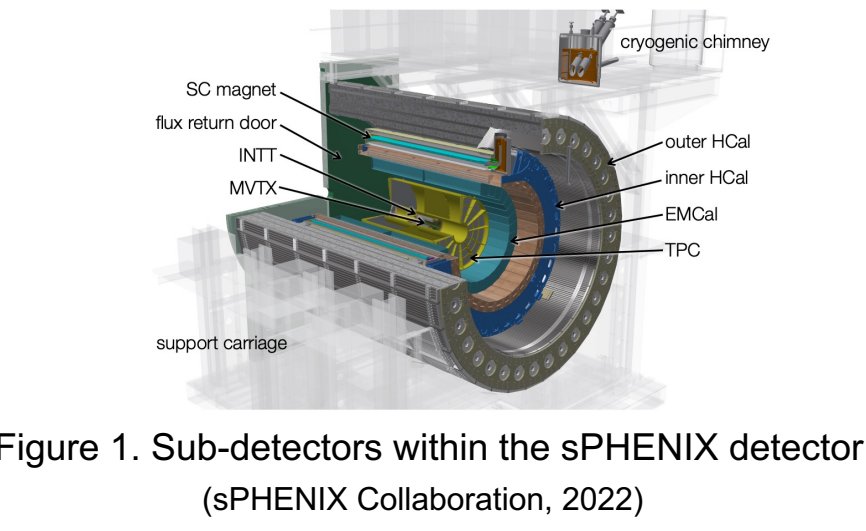


Figure 1. Sub-detectors within the sPHENIX detector (sPHENIX Collaboration, 2022)

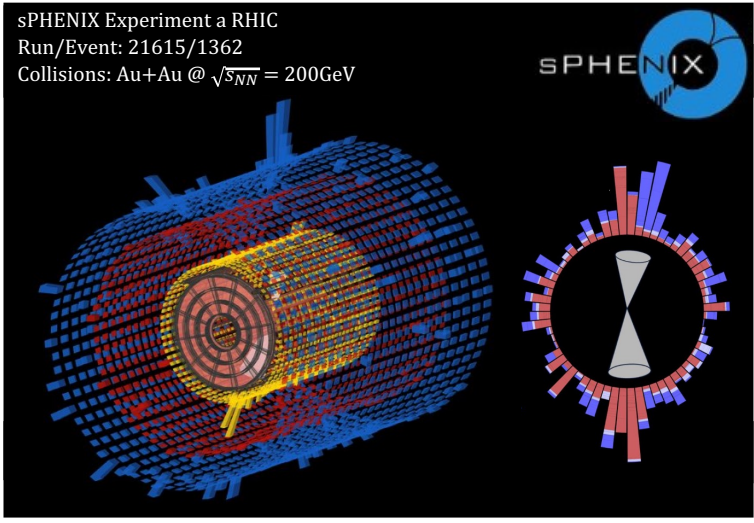


Figure 2. Di-jet observation in the sPHENIX Calorimeters (Ma, 2024)

sPHENIX is composed of 10 sub-detectors, which includes both calorimetry and tracking capabilities (Figure 1). sPHENIX calorimetry includes one electromagnetic calorimeter (EMCAL) and two hadronic calorimeters (iHCal and oHCal). The EMCAL detects particles that interact via the electromagnetic interaction (photons and electrons/positrons), the hadronic calorimeters measure the energy of hadrons, making the calorimeters vital to sPHENIX's jet physics goals. The EMCAL is comprised of 24,576 tungsten-epoxy towers, with 567 Scintillating fibers embedded in each tower. The EMCAL is read out by silicon photomultipliers whose analog output signal is waveform digitized into 12 individual time samples. These waveforms are then fitted with a template to extract the energy, the time and the pedestal of each calorimeter hit.

Calorimeter Waveforms

Waveform fitting

As mentioned above, extracting pedestal, time, and energy values from EMCAL waveforms requires fitting each pulse with a template (Figure 3). Although the fit can accurately extract these values, the large number of EMCAL towers and the unprecedented number of events means that significant computing resources are required to process the data.

Machine learning

To investigate the feasibility of applying machine learning the waveforms and the fit results were used to train a machine learning model. The raw waveforms contain noisy and empty channels (Figure 4a). To simplify the training, cuts to the data were applied to ensure that only high-quality waveforms were used to train the model.

Filtering data

The cuts applied include a cut on the minimum to get rid of empty channels, a cut on the energy (maximum – pedestal) to remove very small signals, and a cut on the first, last, and middle bins to remove noisy channels. This resulted in much cleaner waveforms which were then used to train the model (Figure 4b). The cuts which mainly removed empty channels and resulted in a ~98% reduction in the number of waveforms and we therefore used 10M events to train the model.

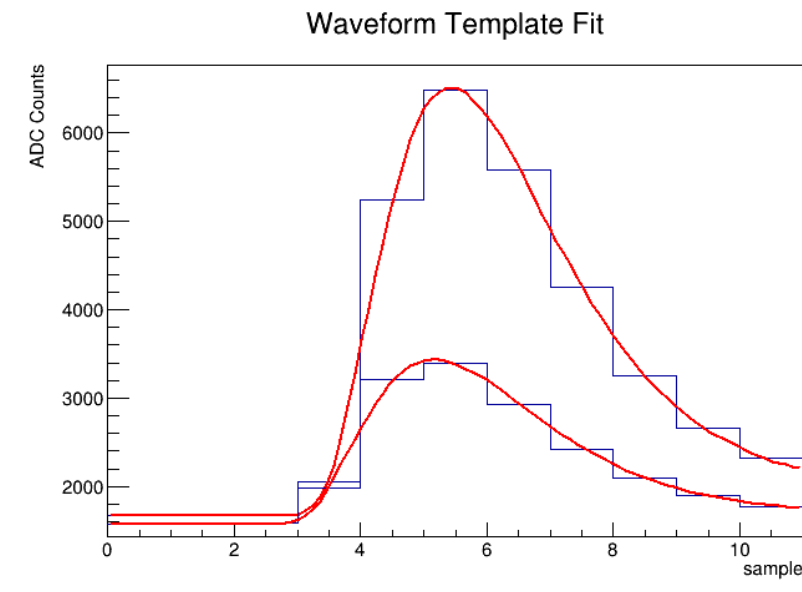


Figure 3. Waveform Histogram with template fit

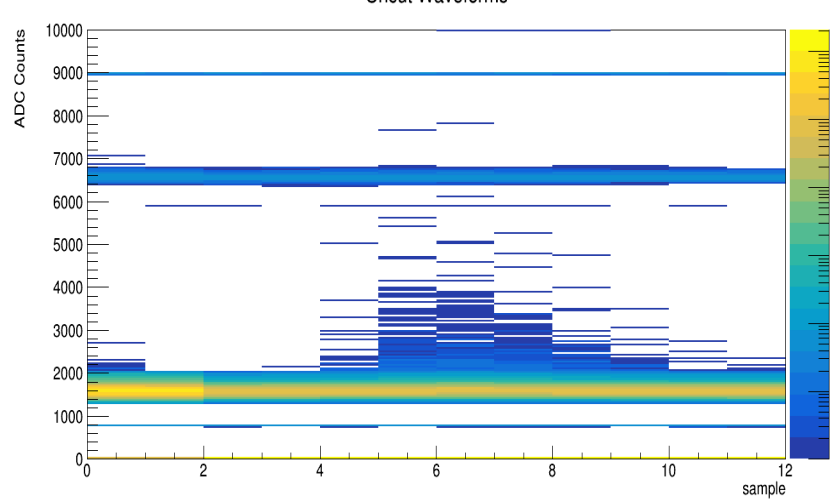


Figure 4a. Waveforms prior to filtering

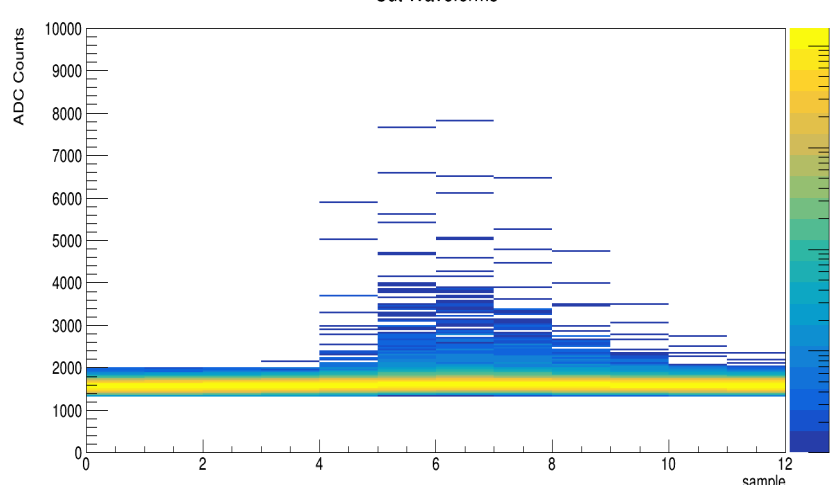


Figure 4b. Waveforms after cuts on amplitude, minimum, and pedestal. This cleaned data was used to train the model

Picture taken by Om Patel and Zen Pinkenburg using ROOT

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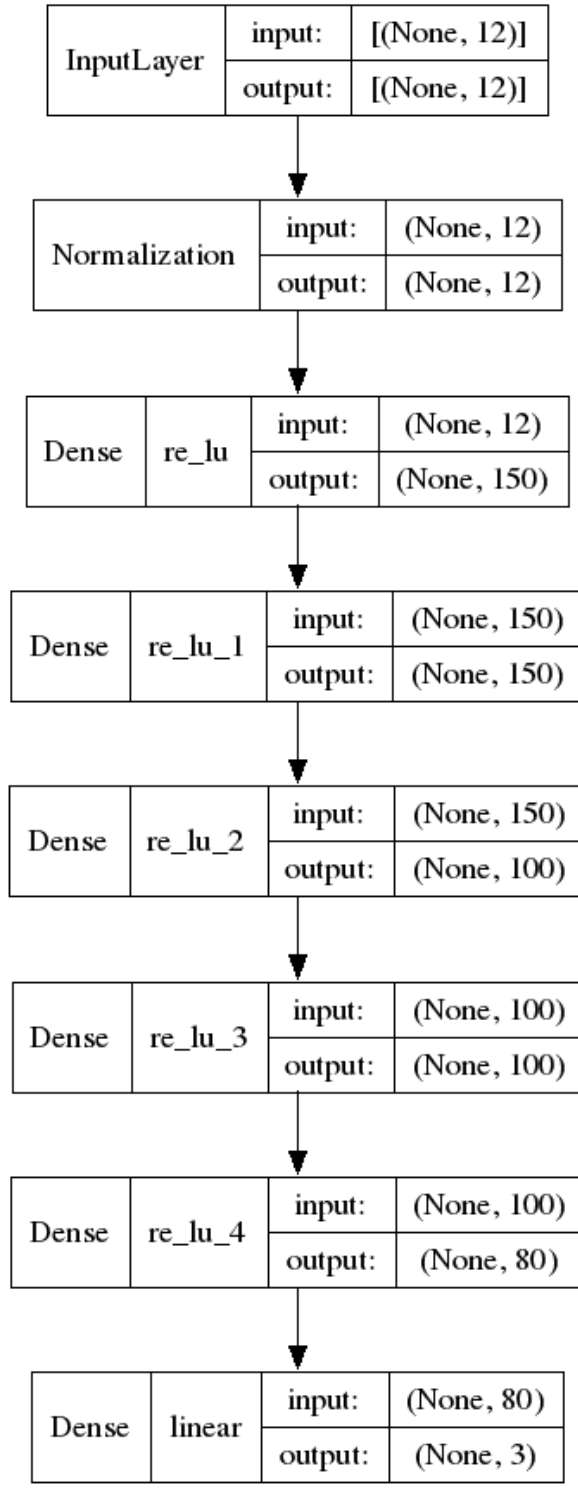
Machine Learning Algorithm

Architecture

The model consists of an input layer, a normalization layer, five dense layers with ReLU activation, and a final dense layer with a linear activation function (Figure 5). The input layer receives the raw data, which consists of 12 features (inputs). The normalization layer standardizes the input data to help the model train more effectively. In the dense layers, each neuron takes inputs from the previous layer, applies weights and a bias, and passes the result through the ReLU activation function. ReLU introduces non-linearity, allowing the network to model more complex patterns. The output of each neuron in one dense layer is then fed forward as input to the neurons in the next layer. The neurons' collective behavior allows the network to learn patterns in the data and make accurate predictions. The final dense layer uses a linear activation function to combine the processed information and returns 3 predicted values. The optimizer used was Adam and the loss function was mean squared error. The machine learning algorithm takes approximately 0.339 ms to predict each fit.

Implementation

Since the model is written in Python and the data is reconstructed by a C++ based framework, a temporary solution was implemented to execute Python code from within the reconstruction. For each event, the waveform values were written to a file, which was then given to the model for predictions. The model returned the predicted energy, time, and pedestal values via another file. To gather sufficient statistics this was run on the sPHENIX 60,000 core condor batch farm.



Picture taken by Om Patel and Zen Pinkenburg using Canva

Figure 5. Architecture of machine learning model

Results and Conclusion

Pedestal, Time, and Energy Predictions

To compare the results, six histograms were generated using ROOT to display the fit-calculated and predicted time, pedestal, and energy values (Figure 6). The histograms have similar shapes, means, and standard deviations, which shows that the model's predictions are very close to the fit's values. However, this only confirms the model's precision, not its accuracy.

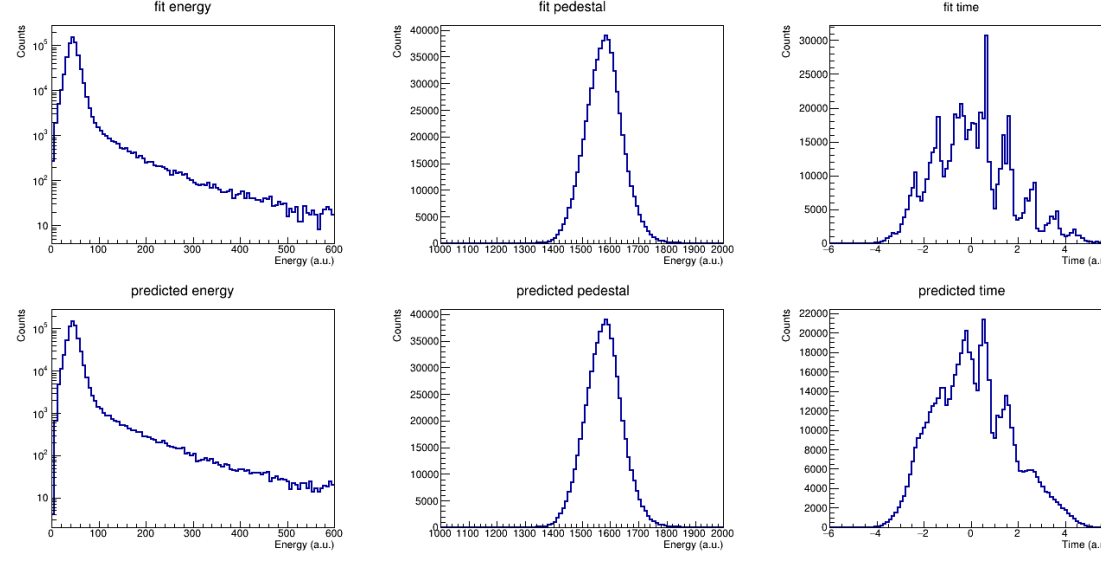


Figure 6: (From left to right) fitted energy, pedestal, and time (top), and predicted energy, pedestal, and time (bottom)

Direct Comparison

When the predicted and fitted energy values are directly compared with each other, they show a strong linear correlation (Figure 7). The statistical errors are smaller than the symbol size. The resulting line of best fit (shown in red) has a slope of 1.04219, which signifies that the model is very precise in the energy predictions (a slope of 1 would mean that the model predicts the energy perfectly). At larger energies, however, the model starts to deviate more from the fitted energy, as shown by the residuals. This is because the model is inaccurate at predicting values outside of its training set, so predicting rare higher energy values is less accurate.

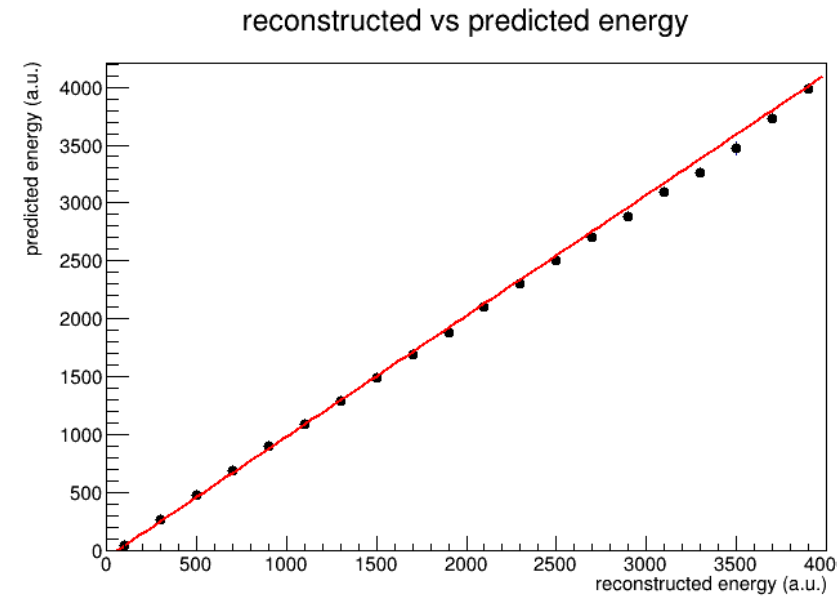


Figure 7: fitted energy (x-axis) and predicted energy (y-axis) with linear regression line in red

Picture taken by Om Patel and Zen Pinkenburg using ROOT

π^0 Reconstruction

A π^0 is a particle that is commonly produced in heavy ion collisions at relativistic energies. When a π^0 decays, it produces two back-to-back photons, whose energies are then measured by the EMCAL. If the two photons hit separate EMCAL towers, the energy produced by the waveform fit of each tower can be used to reconstruct the π^0 s, with the width of the π^0 peak is governed by the energy resolution of the EMCAL. Thus, by using the energy predicted by the model, the π^0 reconstruction provides a direct method to evaluate the accuracy of the model. As seen in Figure 8, the invariant mass distribution is almost identical when using the template fit or the model to extract the tower energies.

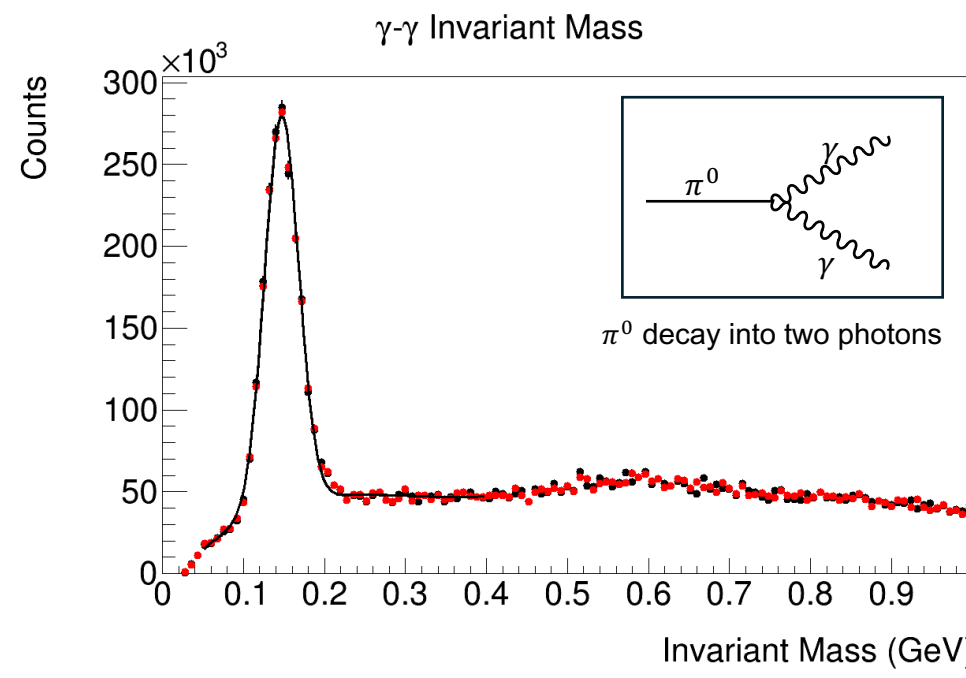


Figure 8. Red represents the invariant mass reconstructed from fitted waveforms and black from waveforms as predicted by the model

Picture taken by Om Patel and Zen Pinkenburg using ROOT

Conclusion

The model was able to make accurate predictions of the energy. But the model's prediction was slower than the waveform fit (averaging over 100 identical events, the machine learning took ~255 ms per event, whereas the fit took ~140 ms). However, the machine learning was not optimized for speed, so it should be possible to decrease the running time significantly by optimizing the inference process. In addition, the implementation of the model into the reconstruction currently requires a restart for every event which takes a large amount of time. Moving to the ONNX software package should allow us to use a C++ API which can then be integrated into the sPHENIX software stack.

Future Research

ONNX

ONNX software package that allows developers to use models and frameworks across different programming languages. Our current method of incorporating the machine learning algorithm within the C++ based sPHENIX reconstruction framework is inefficient. ONNX would resolve this issue by removing the need to open and close the input file in the macros for each event. During the project development, we attempted to use ONNX to convert our model to use it directly in C++. We tried to use several ONNX versions, but each ONNX version has different dependencies and package version requirements, which made configuring each version time-consuming and tedious. We plan a systematic effort to create a working ONNX implementation which can be run on the sPHENIX batch farm.

Hadronic Calorimeters

Since the EMCAL and the hadronic calorimeters use identical readout electronics, it would be very easy for us to analyze digitized waveforms from the hadronic calorimeters. This would be achieved by retraining the machine learning model to predict hadronic signals using the template fit results for the hadronic calorimeters.

Pushing the Model

As of now, our model predicts high-quality waveforms as shown in Figure 4b. So, we plan to test the model in situations where waveform fitting becomes challenging, such as with small energies, non-centered pulses, or unusual patterns caused by readout electronics failures. Ultimately, we aim to have the model predict these non-ideal waveforms and cover ranges where the template fitting approach reaches its limits.



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